

An adaptive seamless phase II/III design for oncology trials with subpopulation selection using correlated survival endpoints[†]

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Although the statistical methods enabling efficient adaptive seamless designs are increasingly well established, it is important to continue to use the endpoints and specifications that best suit the therapy area and stage of development concerned when conducting such a trial. Approaches exist that allow adaptive designs to continue seamlessly either in a subpopulation of patients or in the whole population on the basis of data obtained from the first stage of a phase II/III design: our proposed design adds extra flexibility by also allowing the trial to continue in all patients but with both the subgroup and the full population as co-primary populations. Further, methodology is presented which controls the Type-I error rate at less than 2.5% when the phase II and III endpoints are different but correlated time-to-event endpoints. The operating characteristics of the design are described along with a discussion of the practical aspects in an oncology setting. Copyright © 2010 John Wiley & Sons, Ltd.

Keywords: seamless design; subpopulation selection; phase II/III; oncology; targeted therapy; time-to-event endpoints

1. INTRODUCTION

Adaptive, seamless, phase II/III designs provide an approach whereby important new medicines may be made available to patients more rapidly and efficiently through the utilization of recently developed statistical methodologies. The period between the analysis of phase II data and recruitment of phase III patients, sometimes termed 'white space', can be minimized, while the seamless approach also allows the flexibility to investigate other crucial issues, such as dose finding or subpopulation selection [1], without the need for separate trials. In these studies, there is no break between the learning phase II and the confirmatory phase III and subjects from both phases are used in the final decision-making process. Patient numbers can be used more efficiently as none of the information gained from the learning phase is ignored.

The papers by Schmidli *et al.* [1] and Bretz *et al.* [2] describe a framework for phase II/III clinical trials based on the *p*-value combination methods of Bauer and Kohne [3]. They demonstrate how the closure principle can be applied to test multiple hypotheses under consideration in a treatment selection problem. The testing procedure required to implement the design in any multi-hypothesis situation is now well understood. Brannath *et al.* [4] applied this work more specifically to the issue of adaptive population selection for the development of a targeted therapy and addressed the questions of decision rules and the control of Type-I error.

In this paper, we consider the development of a targeted oncology therapy, where there is an *a priori* hypothesis that a prospectively defined subgroup could particularly benefit. This subgroup may be defined by a biomarker or any pre-defined clinical criteria and it is assumed that the subjects who are members of this subpopulation can be successfully and

reliably identified. The methods of adaptive seamless phase II/III clinical trials will be applied to this setting and extended to allow the most appropriate survival endpoints and decision rules to be employed. Crucially, the experience gained from an intermediate, time-to-event endpoint during the learning phase of the trial can be used to ensure that the confirmatory section of the study includes the most appropriate patient cohorts who may benefit from the treatment, whether this is the entire population or the subpopulation, or whether we continue to test the hypothesis in both groups, here described as the co-primary case.

Broadly speaking, there are two distinct families of seamless design methods in the literature: those based on the *p*-value combination techniques of Bauer and Kohne [3] and those based on the multivariate normal methods of Stallard and Todd [5], which use modified group sequential computations. The *p*-value based phase II/III designs, such as those of Schmidli *et al.* [1] and Brannath *et al.* [4], have been restricted to a single endpoint as combining *p*-values based on differing endpoints would not relate to the test of a sensible hypothesis. Todd and Stallard have extended their work to consider a change of endpoint [6] and Royston *et al.* [7] have addressed related survival outcomes, but both approaches rely on assumptions

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about the joint distributions of test statistics between stages and between subgroups. These methods quickly become complicated when extended to a population selection problem. In this paper, we propose a hypothesis test based on combining test statistics for the final endpoint from each stage, even though the decision to proceed to phase III is based on the intermediate endpoint.

Including a co-primary option in this new design recognizes that there may still be uncertainty after phase II concerning the optimum population to study. Hypotheses relating to both the overall population and pre-defined subpopulation can be investigated using patients from both phases, although some authors have chosen to formulate designs where the subgroup is characterized on an initial stage of patients and examined in a second stage [8]. Wrongly dropping a relevant group at the interim analysis would be costly. Even when an overall effect is found, there can still be interest in the enhanced performance of the subgroup. Furthermore, the trial can be discontinued at the interim analysis if no promising results are seen. A similar situation has been considered by Wang *et al.* [9], although with a single normally distributed endpoint.

The benefits of our proposed design would be lost if the usual intermediate endpoints of an oncology phase II trial could not be incorporated. Although progression free survival (PFS) is beginning to be accepted as a confirmatory endpoint in many cancer types, overall survival (OS) is commonly required in phase III by the regulatory agencies. PFS is frequently used in phase II both for expediency and as it is often the most sensitive endpoint for targeted therapies. We shall extend existing seamless design methods by basing population selection in phase II on PFS but combining OS data from phases II and III in the final analysis, which will support possible licensing claims. In doing this, it is clear that the dependence between PFS and OS must be accounted for. There are two major considerations in the use of PFS alongside OS:

1. The combination of test statistics must test a sensible null hypothesis.
2. The data from two different time-to-event endpoints collected during overlapping periods must be combined without the introduction of bias.

Solutions to these issues and further practical considerations will be presented.

We shall propose a form of seamless phase II/III trial design including population selection based on an intermediate endpoint. Section 2 will outline the trial design and Section 3 will cover the methods for control of Type-I error rate in the multiple testing procedures that are utilized. In Section 4, the specifics of oncology trials and appropriate decision rules for a blinded review are addressed. Section 5 will explain issues raised by the use of associated time-to-event endpoints. The results of simulations of the trial design will be presented in Section 6. The suitability of the trial and the operational issues will be addressed in the closing discussion.

2. THE PROPOSED DESIGN

We shall consider a randomized, parallel group clinical trial with two arms, experimental and control. There will be two distinct stages, an initial learning stage analogous to a randomized phase II trial and a second confirmatory phase analogous to a

randomized phase III trial. An interim analysis takes place based on the first stage subjects only, whereas the final analysis is based on all subjects. Both the full population (F) and a known subgroup (S) are to be investigated for evidence of increased efficacy with the new treatment. The subgroup S should be clearly defined and this definition cannot be altered at the interim analysis.

At the interim analysis, which considers a short-term intermediate time-to-event endpoint, the trial can either:

- Continue in co-primary populations F and S
- Continue in subgroup S only
- Continue in the full population F without an analysis in S
- Stop for futility

Each of the above options has a pre-specified, but potentially different, stage 2 sample size and length of follow-up associated with it. This allows for more subgroup patients to be recruited in the subgroup only case than in scenarios where the full population is continued to be studied.

The stage 1 patients remain in the trial and continue to be monitored for survival events, at the same time as those subjects newly recruited to stage 2, so that this information can be used in the analysis of the long-term endpoint.

The final assessment is of OS for all patients from both stages. However, these are kept as two distinct groups for the purpose of analysis. The appropriate combination methods for integrating OS data from the two stages are described in Section 3. A flow-chart of the proposed design is given in Figure 1.

3. METHODS AND PRINCIPLES

In order to account for the levels of multiplicity involved in the trial, several correction methods and testing procedures must be employed. We shall present methods previously summarized in several papers [1–3,10,11] in the context of our problem.

By considering both the full population F and the subgroup S, multiple one-sided hypotheses about the investigational treatment are taken into account. Within each population, there is a single null hypothesis of no difference in OS between arms. The alternative hypothesis is that the new treatment demonstrates increased efficacy over the comparator in terms of prolonging OS. These null hypotheses for the full population and subgroup will be denoted H_0^F and H_0^S , respectively, and similarly we denote the alternative hypotheses H_1^F and H_1^S . The decisions about which hypotheses to proceed to test will be made at the interim analysis and we discuss rules for doing this in Section 4.

We wish to control at a nominal level α the family-wise error rate, that is, the probability of rejecting at least one true null hypothesis. To do this, we use the closure principle, which involves considering all possible intersection hypotheses $\bigcap H_0^j$, where the H_0^j are in the set of original null hypotheses $\{H_0^F, H_0^S\}$. This produces three hypotheses, H_0^F , H_0^S , and also H_0^{FS} , which specifies that there is no survival difference in either F or S. A null hypothesis H_0^j is only rejected overall if all intersection hypotheses that imply H_0^j are also rejected. Thus, for example, H_0^F can only be rejected overall if individual tests reject both H_0^F and H_0^{FS} at level α .

For subjects recruited in stage $i \in \{1, 2\}$ the p -values for testing H_0^F and H_0^S will be denoted p_i^F and p_i^S , respectively. Specifically, p_1^F and p_1^S are based on the OS data for subjects recruited in stage 1 using their OS through stages 1 and 2, while p_2^F and p_2^S are

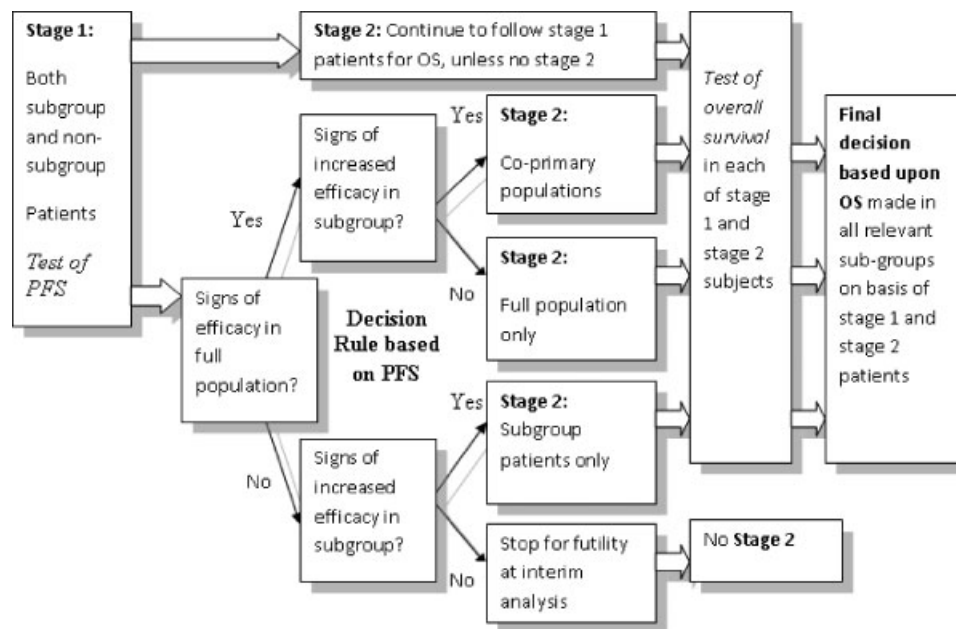


Figure 1. Flow chart of the adaptive seamless design.

calculated from OS data for stage 2 subjects only. The reasons for choosing these definitions and advantages over other possible choices are discussed further in Section 5. The stage i p -value corresponding to H_0^F , p_i^{FS} is a function of p_i^F and p_i^S correcting for multiplicity. We shall use a Hochberg correction [12] with equal weighting of H_0^F and H_0^S , which gives $p_i^{FS} = \min[2 \min\{p_i^F, p_i^S\}, \max\{p_i^F, p_i^S\}]$. Other methods such as that of Dunnett and Tamhane [13] are also possible.

We conduct the final analysis on all subjects using an inverse-normal combination test, which will control the Type-I error rate, regardless of the decision at the interim analysis [14]. Weights w_1 and w_2 , with $w_1 + w_2 = 1$, are specified to combine the p -values from each stage and the null hypothesis is rejected if $C(p_1, p_2) = \{w_1 \Phi^{-1}(1 - p_1) + w_2 \Phi^{-1}(1 - p_2)\} \geq c$. For a one-sided significance level of 0.025, we set $c = 1.96$. The p -values are defined so as to be independent and uniformly distributed under their respective null hypotheses.

An important step in achieving this independence is to *pre-specify* the total length of follow-up of stage 1 subjects for their OS in stages 1 and 2. This can be defined at the start of the trial by setting a calendar time for the end of follow-up or by fixing the total number of failures to be observed among stage 1 subjects. It is not permissible for stage 1 follow-up to be affected by the stage 2 design as decisions about the population being tested and length of follow-up of this population are based on PFS of stage 1 patients, which is liable to be correlated with those subjects' OS. The handling of correlated time-to-event endpoints is discussed further in Section 5 and it is the maintenance of independent test statistics for each stage that allows previous proofs of control of Type-I error via the closure principle [3,10,14] to be applied.

Ideally, the weights w_1 and w_2 would be chosen to be proportional to the square roots of the numbers of OS events observed during each stage. The combination test would then have the attractive property of yielding, approximately, the usual test statistic from a single combined analysis. One might even pre-define different weights in the different hypothesis

Table 1. The weights and p -values to be used in combination tests.

<i>Co-primary case – when considering both H_0^F and H_0^S</i>	
Testing H_0^F :	$w_1 \Phi^{-1}(1 - p_1^F) + w_2 \Phi^{-1}(1 - p_2^F)$
Testing H_0^S :	$w_1 \Phi^{-1}(1 - p_1^S) + w_2 \Phi^{-1}(1 - p_2^S)$
Testing H_0^{FS} :	$w_1 \Phi^{-1}(1 - p_1^{FS}) + w_2 \Phi^{-1}(1 - p_2^{FS})$
<i>F only case – when considering H_0^F only</i>	
Testing H_0^F :	$w_1 \Phi^{-1}(1 - p_1^F) + w_2 \Phi^{-1}(1 - p_2^F)$
Testing H_0^{FS} :	$w_1 \Phi^{-1}(1 - p_1^{FS}) + w_2 \Phi^{-1}(1 - p_2^F)$
<i>S only case – when considering H_0^S only</i>	
Testing H_0^S :	$w_1 \Phi^{-1}(1 - p_1^S) + w_2 \Phi^{-1}(1 - p_2^S)$
Testing H_0^{FS} :	$w_1 \Phi^{-1}(1 - p_1^{FS}) + w_2 \Phi^{-1}(1 - p_2^S)$

tests of H_0^F , H_0^S , and H_0^{FS} to match the likely proportions of events in each stage. However, we are constrained in that weights w_1 and w_2 must be pre-specified and, since the decision at the interim analysis can affect the number of subjects to be recruited in the full population F or subgroup S in stage 2, it is not possible to match the weights to all eventualities.

As a simple compromise solution, we specify weights suited to the important cases of continuing to test for a treatment effect in the full population alone or to test for effects in co-primary populations of the full population and subgroup. Let N_1 and N_2 denote the anticipated numbers of OS events from stage 1 and stage 2 subjects, respectively, in these cases. (The trial design may define the length of follow-up in terms of these numbers or, if follow-up is specified in calendar time, it will be necessary to estimate the numbers of events that will be observed.) We set

$$w_1 = \sqrt{N_1/(N_1 + N_2)} \quad \text{and} \quad w_2 = \sqrt{N_2/(N_1 + N_2)}$$

as the stage 1 and stage 2 weights for the testing of each hypothesis H_0^F , H_0^S , and H_0^{FS} .

To make the final decision, the relevant p -values for H_0^F , H_0^S , and H_0^{FS} are substituted into $C(p_1, p_2)$ as displayed in Table 1. A worked example can be found in Appendix A.

4. INTERIM DECISION RULES

There are several features of oncology clinical trials that are important to incorporate into this design. We have already commented that PFS (the minimum of time to death, time to the growth of existing lesions by a pre-specified amount or time to the occurrence of new lesions [15]) is not yet an acceptable endpoint for confirmatory phase III trials in all cancer types and so the use of a more traditional endpoint may be required for regulatory submissions. However, PFS is more rapidly observed, is often a more sensitive endpoint to drug effects, and in many settings is regarded as either a marker of clinical benefit, a good predictor, or even a valid surrogate for OS. Our design takes advantage of the early availability of PFS in using this endpoint in decision-making at the interim analysis, while retaining OS as the endpoint for final hypothesis testing.

The decision rule to be applied at the interim analysis should be clearly expressed so that the study can be conducted with the sponsor remaining completely blinded to all results at this stage. Possible decision rules could be based on group sequential boundaries or Bayesian methods [4,16]. Whatever approach is used, a simple, unequivocal rule is desirable. In this study, we propose to use a rule based on the estimated hazard ratios for PFS within the full population and the subgroup of interest. Target values are set and the trial only continues in those groups for which the estimated hazard ratio exceeds the target. Simulations of the clinical trial design can be used to choose the thresholds for this decision rule so as to ensure the design has high power to detect an effect in the groups for which the treatment is effective while stopping early for futility if this is appropriate. Table II shows an example where a hazard ratio less than 1 indicates increased benefit from the experimental treatment.

The possibility to stop for futility at the interim analysis is valuable for sponsors and investigators to optimize the investment of resources. Futility stopping will decrease the trial's Type-I error rate since there is less chance of continuing past the interim analysis and so a reduced probability of rejecting a null hypothesis erroneously at the end of the trial. Similarly, power is also decreased, as there are two opportunities to stop without rejecting a null hypothesis.

We make no attempt in our design to 'recover' the reduction in Type-I error probability due to stopping for futility. This would run the risk of the error rate increasing if a decision were made to continue despite failing a futility rule. Such an approach is in keeping with the increasing use of 'non-binding' futility boundaries. In any case, an adjustment to recover Type-I error probability would be hampered by the need to estimate the correlation between endpoints. Our approach guarantees that the Type-I error rate is fully controlled in the proposed design. Simulations of the effect of the chosen decision rule will be seen in Section 6.

Table II. Example decision rule.		
PFS hazard ratio estimates at the interim analysis	$\hat{HR}^F < 0.8$	$\hat{HR}^F \geq 0.8$
$\hat{HR}^S < 0.6$	Continue with co-primary analysis	Continue in subgroup S only
$\hat{HR}^S \geq 0.6$	Continue in full population F only	Stop for futility

5. TIME-TO-EVENT ENDPOINTS

As discussed in the previous section, it is important in practical terms that separate short-term and long-term time-to-event endpoints can be used in different aspects of a seamless design. However, Bauer and Posch [17] have identified a problem in using time-to-event endpoints in a two-stage adaptive design. If the choice of stage 2 population is affected by short-term responses of stage 1 subjects and follow-up of the long-term response of these stage 1 subjects contributes to the stage 2 log-rank statistic for this endpoint, then this statistic may not have the desired null distribution. Our solution is to perform a trial with the following characteristics:

Option A: Our Solution.

- PFS from stage 1 subjects is used at the interim analysis in deciding which subgroups to follow, but does not contribute to the final test statistics.
- p -Values for comparisons of OS based on log-rank tests are produced separately for those subjects recruited to stage 1 and those recruited to stage 2. In particular, the additional follow-up of stage 1 subjects during stage 2 contributes to stage 1 p -values. Also, the extent of total follow-up of stage 1 subjects for OS is pre-specified and, hence, not affected by their PFS times observed at the interim analysis.
- The comparisons of OS are not carried out until the end of the trial, when a pre-specified number of events or length of follow-up is reached for each set of stage 1 and stage 2 patients, respectively. At this point, the p -values from each stage are combined.

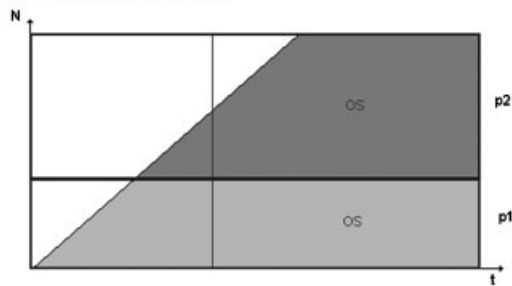
This method ensures that a selected hypothesis can be tested with proper protection of the family-wise Type-I error rate. Our construction avoids the problem noted by Bauer and Posch [17] because we allocate stage 2 follow-up of stage 1 recruits to the stage 1 p -values, p_1^F , p_1^S and p_1^{FS} , rather than their stage 2 equivalents. This guarantee of independent test statistics ensures that existing theory regarding control of Type-I error [2,3,10,14] continues to hold under the selection of subpopulations, as has been previously described in similar designs without an intermediate endpoint [1,4].

The potential dangers in defining a procedure for our situation are subtle and we shall illustrate them by describing, in addition to our proposed approach, option A, a series of design options B to D which are unsuitable for reasons of ambiguity, inefficiency or bias. Figure 2 illustrates how the different phases of follow-up on PFS and OS for stage 1 and stage 2 recruits are used in each option. As a point of reference, in our proposed method, PFS outcomes of stage 1 subjects are used to make the interim decision on which populations to follow. Then, the p -values p_1^F , p_1^S and p_1^{FS} are calculated from OS results for stage 1 subjects using the information about these subjects available at the end of the study. The p -values p_2^F , p_2^S and p_2^{FS} are calculated using the OS data accruing in stage 2 for the new stage 2 recruits.

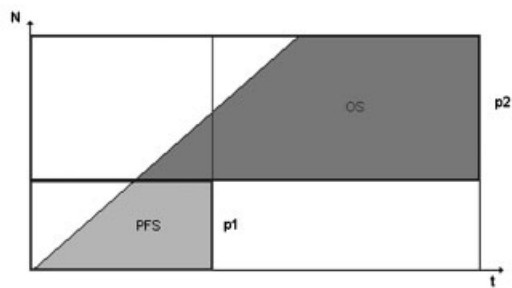
Option B: PFS results from stage 1 subjects are used to define the p -values p_1^F , p_1^S and p_1^{FS} at the interim analysis. OS of stage 2 recruits provides the second set of p -values, p_2^F , p_2^S and p_2^{FS} . The p -values from the two stages are then combined in an inverse-normal combination test.

There is ambiguity here as it is questionable what hypothesis is finally tested in the combination test. Combining stage 1 p -values based on PFS and stage 2 p -values from OS will not provide a clear and sensible conclusion.

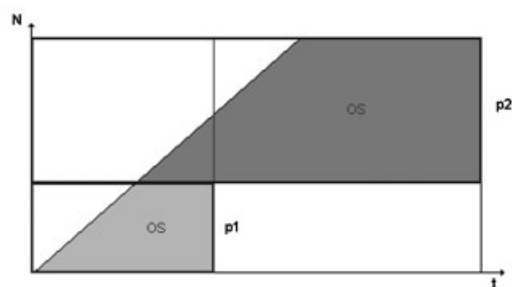
Option A: Our proposed solution:



Option B:



Option C:



Option D:

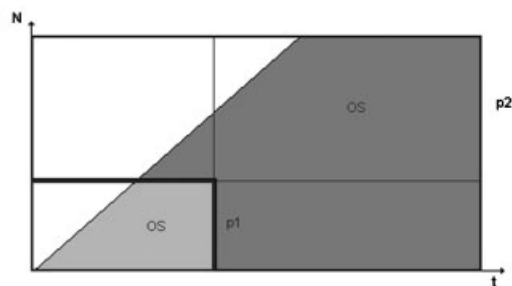


Figure 2. Options for the combination of stage 1 and 2 survival data. Diagrammatic representation of design options. N =patient numbers, t =time. The horizontal reference line is at the number recruited to stage 1. The vertical reference line is at the time of the interim analysis. The shaded box indicates the set of subjects and associated OS or PFS data used to calculate each p -value.

Option C: PFS outcomes of stage 1 subjects are used to make the interim decision on which populations to follow. The p -values p_1^F, p_1^S and p_1^{FS} are calculated from OS results on stage 1 subjects at the time of the interim analysis, while p_2^F, p_2^S and p_2^{FS} are produced from OS results at end of stage 2 solely for the new subjects recruited in stage 2.

Although this solution is mathematically sound (as no bias is introduced) it suffers from practical problems. Of the stage 1 subjects, only those deaths observed by the time of interim analysis contribute to the final hypothesis test. Very few events

can be expected by this time and hence overall power will be lower than it could be.

Option D: PFS outcomes of stage 1 subjects are used to make the interim decision on which populations to follow and p -values p_1^F, p_1^S and p_1^{FS} are calculated from OS results on stage 1 subjects at the time of the interim analysis. The stage 2 p -values p_2^F, p_2^S and p_2^{FS} are calculated using all additional OS data accruing in stage 2, both from the new stage 2 recruits and from continued follow-up of the OS of stage 1 subjects.

The intention behind this option is to make efficient use of the stage 1 subjects by including their OS events that did not contribute to p_1^F, p_1^S and p_1^{FS} in the stage 2 p -values instead. Wassmer [18] describes how independent increments arise in log-rank statistics based on increasing follow-up of a given time-to-event endpoint. Applying this independence property, the stage 2 p -values are calculated from the weighted increment of the log-rank statistics based on the total follow-up of all subjects minus that based only on events prior to the interim analysis.

However for this approach to be valid when multiple endpoints are used, the future increment in the OS log-rank statistic for stage 2 follow-up must be independent of the PFS data used to select hypotheses for testing in stage 2. This may not be the case as there is a set of subjects who have progressed by the interim analysis, but not yet died. These subjects contribute to both the interim decision and to p_2 and so introduce bias as they have a bearing on both which group is chosen and the final result derived from that group.

Suppose, for example, there is a positive correlation between outcomes PFS and OS, so patients with better than average PFS can also expect good OS. Then, under the null hypothesis H_0^S , the decision to follow subgroup S will be taken when those members of S on the experimental treatment show better PFS than the controls, and so follow-up of OS for these stage 1 subjects is also likely to produce better outcomes for the experimental group. Deciding to test H_0^S when rejection of H_0^S is more likely will bias the test of this hypothesis. This bias and the resulting contamination of the stage 2 p -values is seen very clearly in the example we present in Appendix B. Thus, option D is not viable, in keeping with the warnings of Bauer and Posch [17] on adaptations based on time-to-event endpoints and correlated surrogates.

Returning to our proposed design (Figure 2, image 1), since there are clearly separate sets of subjects involved in stage 1 and 2 p -values, there is no chance of bias being carried across stages. This solution is based on two pieces of insight. First, a seamless design does not have to divide data on OS into deaths occurring before and after the interim analysis: rather, the subdivision can be into the survival results of any two well-defined, non-intersecting patient groups. Second, as an intermediate endpoint is used during the interim analysis, the calculation of p -values based on OS for stage 1 subjects can be deferred until the end of the trial, allowing long-term follow-up of all stage 1 subjects to contribute to the log-rank statistic for OS.

In the interests of operational effectiveness, it is recommended that recruitment should be allowed to continue without cessation at the interim analysis. The sample sizes and number of events required to trigger OS analysis can be pre-specified for each group, allowing appropriate weights to be set for each stage. Once the required number of subjects has been entered into stage 1 recruitment can switch immediately to stage 2 while possible additional follow-up for PFS of stage 1 subjects and data cleaning for the interim analysis takes place.

Any patients recruited after this switch would be considered as part of stage 2 and would not be included in the PFS interim analysis. This approach gives the maximal time savings, but may risk additional costs if subjects are recruited who are not then required for stage 2, e.g. when recruitment continues in the full population, but it is decided at the interim analysis to restrict attention to the subpopulation.

One could alternatively define stage 1 to include all the subjects recruited up until the interim analysis. This would include subjects with immature PFS data at the time of the interim analysis and, as it is not uncommon for survival curves to take time to separate, these late recruits would contribute little to the PFS comparison and they may even dilute the magnitude of any treatment effects on PFS. Consequently, our preference is for recruitment to switch to stage 2 some time before the interim analysis.

The final analyses for each stage can be triggered by the occurrence of a pre-set number of OS events or, alternatively, by reaching a specified calendar time. Although it might seem more natural for the follow-up of OS for stage 1 and stage 2 patients to conclude at the same time, the validity of the statistical test relies on the length of follow-up of stage 1 subjects being pre-set at the beginning of the study. It should be noted that this difference in cut-off points means that there may be events occurring in one of the two cohorts (of stage 1 or stage 2 patients) which must be disregarded from the final analysis if they occur in the first cohort to reach this cut-off, but in the period in between this time and the second cohort completing. Hence, for example, if the stage 1 patients reach their cut-off first, additional events in this cohort, which occur while follow-up of stage 2 patients is completed, will not be used in the final analysis and vice versa.

6. SIMULATION RESULTS

The effectiveness of a seamless design depends on many parameters, including event rates, recruitment rates, the size of the subgroup, and the hazard ratios for PFS and OS in each group. It is therefore necessary to simulate possible designs

under a variety of conditions in order to evaluate their operating characteristics. The simulation results presented in Tables III, IV and V demonstrate good performance of our proposed methods in a range of situations. In these examples, correlated OS and PFS times with marginal exponential distributions were defined using a mixture approach [19]. Study entry was modeled with a univariate distribution for arrival times.

Table III demonstrates that, in agreement with theory, the Type-I error rate is controlled at less than 2.5%. This is true both for testing of the intersection hypothesis, and also for the global family-wise error rate. This overall error rate is lower than the intersection hypothesis rejection rate as on some occasions H_0^{FS} will be rejected, but not H_0^F , or H_0^S . These results cover a wide range of values for the correlation between endpoints in the case where there is no futility stopping and the trial always continues to the co-primary scenario. The absence of early stopping for futility means that the conservatism seen here is due to the corrections for multiplicity. If futility stopping had been included here then this would have reduced the Type-I error rate. The independence between first and second stage p -values is also demonstrated with the magnitude of all correlations being less than 0.05.

Table IV gives estimated probabilities of the various interim and final decisions for our procedure under three different decision rules. The interim analysis occurs after 200 PFS events from the 300 patients recruited to stage 1. A further 800 patients are then recruited to stage 2 if we continue in the full population or co-primary case, with 400 recruited if only the subgroup is used in stage 2. The final analysis of OS is performed when 250 deaths have occurred in stage 1 and stage 2 subjects have produced 500 deaths in the full or co-primary case or 250 deaths in the subgroup only case. The subgroup is one fourth of the population and there is a correlation of 0.7 between PFS and OS. The results show that if the trial continues to stage 2, H_0^{FS} is almost always rejected for the two cases considered. However, stopping for futility at the end of stage 1 occurs in 5–25% of cases and this rate is highly dependent on the decision rule chosen, as would be expected. As the conditions on estimated hazard ratios for PFS become more stringent from decision rule

Table III. Control of type-I error under H_0 .					
Correlation between OS and PFS	Family-wise Type-I error (rejection of H_0^F or H_0^S)	Probability of rejection of H_0^{FS}	Correlation between p_1^F and p_2^F	Correlation between p_1^S and p_2^S	Correlation between p_1^{FS} and p_2^{FS}
0.1	0.018	0.020	0.009	−0.020	−0.018
0.2	0.019	0.021	−0.019	−0.013	−0.008
0.3	0.022	0.024	0.005	−0.031	−0.037
0.4	0.024	0.026	0.020	−0.010	0.013
0.5	0.020	0.023	−0.011	0.016	0.008
0.6	0.019	0.021	−0.019	−0.027	−0.014
0.7	0.021	0.023	0.012	0.005	0.009
0.8	0.020	0.023	0.025	0.045	0.052
0.9	0.021	0.023	0.000	0.006	0.017
Each row is based on 3000 simulations of a trial under the global null hypothesis where all hazard ratios are equal to one. A family-wise error is defined as rejecting at least one of the true hypotheses H_0^F and H_0^S in the closed testing procedure. Three hundred patients were simulated to be recruited to stage 1 and 800 to stage 2. The interim analysis occurs after 200 progressions. The final analysis occurs after 250 deaths in stage 1 subjects and 500 deaths in stage 2 subjects. The full correction methods are used, with the exception of stopping for futility. The standard error for estimated type-I error rates is approximately 0.003 based on the normal approximation to the binomial.					

Table IV. Expected proportions of trial outcomes for three decision rules.

Three rules based on estimated hazard ratios for PFS were applied at the end of the first stage:

Decision rule 1: Continue in the full population if $\hat{HR}^F < 0.9$ and continue in the subgroup if $\hat{HR}^S < 0.7$; if both apply continue with co-primary populations; if neither apply stop for futility.

Decision rule 2: Continue in the full population if $\hat{HR}^F < 0.8$ and continue in the subgroup if $\hat{HR}^S < 0.8$; if both apply continue with co-primary populations; if neither apply stop for futility.

Decision rule 3: Continue in only the full population if $\hat{HR}^F < 0.8$ and continue in the subgroup if $\hat{HR}^S < 0.6$; if both apply continue with co-primary populations; if neither apply stop for futility.

Stop for futility	Continued in co-primary population				Continued in F		Continued in S		Total power Reject either H_0	
Don't reject any H_0	Don't reject any H_0	Reject H_0^F & H_0^S	Reject H_0^F only	Reject H_0^S only	Don't reject H_0	Reject H_0^F only	Don't reject H_0	Reject H_0^S only		
Decision rule 1										
Case 1) Hazard ratios: $HR^{Sc} = 0.8$, $HR^S = 0.6$, for both PFS and OS	5.5%	0.9%	64.6%	2.8%	0.5%	0.7%	22.2%	0.1%	2.8%	92.9%
Case 2) Hazard ratios: $HR^{Sc} = 0.9$, $HR^S = 0.5$, for both PFS and OS	10.5%	0.3%	59.4%	0.3%	4.5%	0.7%	10.6%	0.0%	13.8%	88.5%
Decision rule 2										
Case 1) Hazard ratios: $HR^{Sc} = 0.8$, $HR^S = 0.6$, for both PFS and OS	10.2%	0.4%	58.4%	3.3%	0.4%	0.1%	5.9%	0.1%	21.3%	89.2%
Case 2) Hazard ratios: $HR^{Sc} = 0.9$, $HR^S = 0.5$, for both PFS and OS	11.2%	0.2%	46.9%	0.2%	2.6%	0.1%	2.9%	0.0%	35.9%	88.4%
Decision rule 3										
Case 1) Hazard ratios: $HR^{Sc} = 0.8$, $HR^S = 0.6$, for both PFS and OS	25.1%	0.4%	37.0%	2.2%	0.5%	0.6%	24.0%	0.1%	10.1%	73.8%
Case 2) Hazard ratios: $HR^{Sc} = 0.9$, $HR^S = 0.5$, for both PFS and OS	23.8%	0.1%	39.6%	0.1%	1.8%	1.0%	9.1%	0.0%	24.4%	75.1%
The table shows the proportion of simulations in which each continuation decision and final conclusion were reached. These are estimates of overall, rather than conditional, probabilities. Where rows do not sum to 100%, this is due to rounding errors. Each case is based upon 3000 simulations of a trial with a subgroup of one-fourth of the total population, 300 subjects recruited in stage 1, 800 in stage 2 in the full and co-primary cases, and 400 in the subgroup only case. The interim analysis occurs after 200 progressions. The final analysis occurs after 250 deaths in stage 1, 500 deaths in stage 2 for the full or co-primary cases, and 250 deaths in stage 2 for the subgroup only case. The median PFS is 4 months, the median OS is 12 months, and the correlation between these is 0.7. Stage 1 recruitment was modelled uniformly over 12 months and stage 2 uniformly over 18 months. We denote the hazard ratio in the subgroup by HR^S and that in the complement of this group by HR^{Sc} .										

1 through to decision rule 3 the likelihood of stopping for futility increases. This illustrates how the choice of decision rule is critical and how simulation can aid in the design process.

Table V examines our procedure's operating characteristics for various alternative hypotheses in a trial with the same sample size as in Table IV. Criteria $\hat{HR}^F < 0.9$ and $\hat{HR}^S < 0.7$ are applied for continuing in stage 2 with the full population and the subgroup respectively, as in decision rule 1 of Table IV. The simulations demonstrate that correct conclusions are reached sufficiently often in the various scenarios. As one would expect, the relative size of the subgroup has a bearing on the outcomes.

In scenario 1, where all null hypotheses hold, the Type-I error is maintained below 2.5%, at around 1%. The decrease relative to the Type-I error rates seen in Table III is due to stopping for futility on 70–80% of occasions, so the trial would not continue with the costly second phase too frequently. Making stage 1 larger while retaining the same criteria for $\hat{HR}^F$ and $\hat{HR}^S$ would increase this probability of stopping still further.

In scenario 2, where there is a difference in survival in the subgroup only, the relative size of the group has a clear impact.

As the size of the subgroup increases, this increases the overall treatment effect and so increases the rate of rejection of H_0^F as well as H_0^S . Rates of stopping for futility or continuing in the subgroup only also drop as a larger proportion of patients benefit. In order to have a chance of finding a treatment effect in S only, there must be a large survival difference compared to F and a reasonably small subgroup. For larger subgroups, efficacy is likely to be demonstrated in both S and F, as a higher proportion of patients benefit from the treatment.

In scenario 3, where all subgroup and non-subgroup patients show a difference in survival between the experimental treatment and control, there is a high power to reject H_0^{FS} . The effect in F is reliably detected, while it is rare for only H_0^S to be rejected without also rejecting H_0^F . When the trial does continue to a second stage it is in F alone on around half of these occasions, although when the co-primary option is taken H_0^S is also generally rejected unless the subgroup is small. The results in scenario 4 are similar to those in scenario 3, with a moderate treatment effect for the non-subgroup patients, but a larger effect in the subgroup patients. Although there is little chance of rejecting H_0^S in the subgroup only, in contrast to scenario 3

Table V. Expected proportions of trial outcomes in various scenarios using decision rule 1.

Subgroup incidence Proportion of patients in S	Stop for futility Don't reject any H_0 (%)	Continued in co-primary population				Continued in F		Continued in S		Total power
		Don't reject any H_0 (%)	Reject H_0^F & H_0^S (%)	Reject H_0^F only (%)	Reject H_0^S only (%)	Don't reject H_0 (%)	Reject H_0^F only (%)	Don't reject H_0 (%)	Reject H_0^S only (%)	Reject either H_0 (%)
Scenario 1) Hazard ratios: $HR^{Sc} = 1$, $HR^S = 1$ for both PFS and OS										
0.2	70.0	6.3	0.1	0.2	0.3	15.5	0.4	6.9	0.3	1.3
0.4	75.9	3.8	0.1	0.1	0.1	17.9	0.8	1.3	0.1	1.1
0.6	75.6	2.7	0.2	0.0	0.1	20.4	0.8	0.3	0.0	1.0
0.8	76.6	1.3	0.2	0.0	0.0	21.0	0.9	0.0	0.0	1.1
Scenario 2) Hazard ratios: $HR^{Sc} = 1$, $HR^S = 0.6$ for both PFS and OS										
0.2	23.1	7.1	14.3	0.4	18.3	6.5	3.0	0.6	26.7	62.7
0.4	13.7	0.9	53.3	0.0	9.9	2.4	8.7	0.0	10.9	82.9
0.6	6.3	0.0	78.0	0.0	0.7	0.3	13.7	0.0	0.9	93.3
0.8	2.2	0.0	81.9	0.0	0.0	0.0	15.9	0.0	0.0	97.8
Scenario 3) Hazard ratios: $HR^{Sc} = 0.7$, $HR^S = 0.7$ for both PFS and OS										
0.2	3.0	0.1	35.2	15.9	0.0	0.2	45.0	0.1	0.4	96.6
0.4	3.7	0.1	44.4	4.0	0.0	0.2	47.5	0.0	0.2	96.0
0.6	3.3	0.2	48.7	0.6	0.0	0.2	46.8	0.0	0.0	96.2
0.8	4.0	0.0	49.0	0.2	0.0	0.4	46.5	0.0	0.0	95.6
Scenario 4) Hazard ratios: $HR^{Sc} = 0.7$, $HR^S = 0.5$ for PFS, $HR^{Sc} = 0.8$, $HR^S = 0.6$ for OS,										
0.2	0.7	1.9	72.8	7.6	1.1	1.0	14.5	0.0	0.5	96.4
0.4	0.3	0.3	91.7	0.6	0.2	0.1	6.7	0.0	0.1	99.3
0.6	0.1	0.0	96.1	0.0	0.0	0.0	3.8	0.0	0.0	99.9
0.8	0.0	0.0	98.3	0.0	0.0	0.0	1.7	0.0	0.0	100.0
Scenario 5) Hazard ratios: $HR^{Sc} = 2$, $HR^S = 0.5$ for PFS, $HR^{Sc} = 1.7$, $HR^S = 0.6$ for OS,										
0.2	18.3	0.0	0.0	0.0	0.0	0.0	0.0	3.0	78.7	78.7
0.4	9.1	0.2	0.0	0.0	3.7	0.0	0.0	0.6	86.4	90.1
0.6	4.5	0.0	21.0	0.0	40.7	0.0	0.0	0.0	33.8	95.5
0.8	0.7	0.0	97.1	0.0	0.9	0.0	1.0	0.0	0.2	99.3
The table shows the proportion of simulations in which each continuation decision and final conclusion were reached. These are estimates of overall, rather than conditional, probabilities. Each row of results is based upon 3000 simulations of a trial with a subgroup of one fourth of the total population, 300 subjects recruited in stage 1, 800 in stage 2 in the full and co-primary cases and 400 in the subgroup only case. The interim analysis occurs after 200 progressions. The final analysis occurs after 250 deaths in stage 1, 500 deaths in stage 2 for the full or co-primary cases and 250 deaths in stage 2 for the subgroup only case. The median PFS is 4 months, the median OS is 12 months and the correlation between these is 0.7. Decision rule 1 from table 4 is used with criteria $\hat{H}R^F < 0.9$ and $\hat{H}R^S < 0.7$ for continuing with the full population and the subgroup. Stage 1 recruitment was modelled uniformly over 12 months and stage 2 uniformly over 18 months. We denote the hazard ratio in the subgroup by HR^S and that in the complement of this group by HR^{Sc} .										

the most common result is to continue to the co-primary case and to reject both H_0^S and H_0^F , and this is the desirable outcome.

The final situation considered is scenario 5, where the treatment effects on the subgroup and non-subgroup patients are in opposing directions, with a greater effect on PFS than OS. One would hope in this case to reject H_0^S , but H_0^F should not be rejected. Table V shows that, indeed, when the subgroup is small the trial virtually never continues to test H_0^F in the second stage and H_0^S is often rejected. As the size of this subgroup increases, the treatment effect for the full population is enhanced and there is a tendency to continue to stage 2 with co-primary populations. As the subgroup increases further, first H_0^S alone is rejected and then, when this group dominates, both H_0^S and H_0^F are rejected regularly.

It is of interest to draw comparisons with more conventional development programs, with a separate phase II and III. Suppose conventionally 250 subjects are observed in phase II and, on the basis of their PFS and OS outcomes, it is decided whether to proceed with phase III. If so, 500 subjects are recruited to phase III and followed until 331 OS events are observed. Conditional on proceeding to the phase III trial, this design has 90% power to detect a hazard ratio of 0.7 in the full population when applying a one-sided test of 2.5% significance.

We can contrast this with our proposed design where subgroup and non-subgroup patients benefit similarly from the compound, with hazard ratios of 0.7 in both groups. With no futility stopping, an interim PFS analysis of the 250 stage 1 patients could be made after 200 progressions with the eventual

survival analysis occurring after 227 deaths if the trial continued into stage 2. This stage 2 would be similar to the conventional phase III trial, with 500 subjects and a survival analysis after 331 events. This design has 97% power of rejecting at least one null hypothesis; (55% of the time we reject H_0^F and H_0^S and 42% of the time we reject H_0^F only). Therefore the re-use of the stage 1 patients in the OS analysis has increased the power, which in both cases is stated conditionally on proceeding to stage 2.

If we had run a similar population selection process in stage 1, but only used the stage 2 survival events in the final analysis, so a more analogous approach to separate phase II and III trials, then the program would have had just 85% power; (35% of this time we reject H_0^F and H_0^S and 50% of the time reject H_0^F only). The drop in power from 90% is due in part to conservatism in the multiple testing procedure, but the overall loss of power demonstrates the benefit of utilizing the stage 1 patients. In fact, without futility stopping, a sample size of 400 events for our proposed design (for example, split 175 and 225 between stages 1 and 2) would be sufficient to give 90% power of detecting a hazard ratio of 0.7 in our proposed new design, using a one-sided test of 2.5% significance.

If futility stopping is introduced, then the unconditional power of each development program across phases II and III can be compared. In this study, identical decision rules, with hazard ratio boundaries for PFS at 0.9 for F and 0.75 for S, could be introduced to both our proposed design and to the end of phase II decision in the conventional case. We now find that the unconditional power drops to 95% for our proposed design but to 84% for the conventional program.

However, the adaptive design's main potential benefits are the savings in time and money that will be achieved in other situations where there is a substantial subgroup difference. The overall number of subjects used in the development program could well be lower under an adaptive approach as the need for a prior phase II trial or concurrent subpopulation trial is removed.

7. DISCUSSION

The statistical methodology for adaptive seamless phase II/III designs allows one to conduct efficient trials that can simultaneously investigate several vital questions for drug development, including the identification of the most beneficial subgroup for a new treatment. We have extended the previous work on the subject to show how the trial design can be modified to suit the particular needs of oncology studies. We have developed the flexible use of a co-primary population based on intermediate time-to-event endpoints and have demonstrated that these modifications do not inflate Type-I error. The most important aspect of these designs is that they perform capably in realistic situations, as confirmed by the simulations in Table V. Such simulations help to identify situations where the design is most suitable, for example where the difference in treatment effects between the subgroup and non-subgroup subjects is considerable and where this group accounts for a relatively small proportion of the total population.

Any novel adaptive design for subgroup selection, for example Freidlin and Simon's alternative designs [8,20], must address these vital issues of the control of Type-I error and the exclusion of bias. The 'adaptive signature design' [8] uses an exploratory phase to identify molecular classifiers and hence stage 1 data cannot be used in the primary analysis of the subgroup hypothesis. Our

design is applicable when a possible predictive biomarker is already identified and this allows it to incorporate data from both stages in the primary hypothesis test. Hung *et al.* [21] are critical of the combination of an exploratory stage based on surrogate endpoints with a confirmatory stage, but in our design all hypotheses are fixed and pre-defined and so in the first stage we learn about proposed hypotheses rather than generate them. Our planned use of multiple endpoints is in contrast to a mid-trial adaptation that would change the hypotheses under investigation, as has been discussed by Jennison and Turnbull [11] and Hung *et al.* [22].

Adaptive designs have the potential to make drug development more efficient, although this can be at a price of increased complexity, logistical challenges for drug supply and other operational issues. The large amount of intensive planning that must be undertaken to define go/no-go criteria is likely to delay the start of the trial. The sponsor must also have identified the necessary resources should the trial continue to the completion of the phase III portion. Should the trial terminate at the interim analysis then there is likely to have been a larger investment compared with a conventional phase II study. This is due to the advance recruitment of some of the phase III subjects and the earlier phasing of costs required for commercial formulation development and manufacture.

One of the first questions that should be addressed when considering a seamless phase II/III design is the need for short-term responses of stage 1 patients to be observed, validated and analyzed quickly in comparison with the recruitment rate for stage 2 so that a timely decision can be made before large phase III costs begin to be accrued. Unless a rapid endpoint is available, this may preclude the use of the design.

An important consideration is whether the sponsor should remain blind to the stage 1 results. Some authors argue this may be achieved by a firewall [23], although it is a generally accepted principle that access to interim analyses from pivotal trials should be restricted to the Independent Data Monitoring Committee. We believe that if the sponsor pre-defines and fully describes the criteria to be applied at the end of phase II, then it should be possible to remain blind to accumulating results. Passing the threshold for continuation to phase III implies the data are promising, but the sponsor has no need to know just how good they are.

Our proposed design should only be applied if there is an *a priori* hypothesis about increased efficacy in a defined subpopulation of patients. Further, this subgroup must be well characterized, e.g. by a biomarker, and any quantitative thresholds classifying membership of the subgroup must be fully pre-specified. This definition must be made prior to the phase II analysis, and would ideally be specified prior to the trial commencing. In addition, a reliable and reproducible assay must have been developed in order to avoid any dilution of the benefits of the design. Any additional knowledge on the proportion of subjects who belong to the subgroup and the relative effects expected in each population could further aid in the planning of the trial through simulations such as those reported in this paper.

This approach could easily be applied in other therapy areas and to endpoints other than time-to-event. The closure principle and multiplicity correction procedures extend to larger numbers of subgroups and so it is feasible to study more than the single subgroup that was investigated in this study. However, if more than one group were taken forward to the second stage or if

there were complicated relationships and interdependencies between groups then this would introduce further levels of complexity. The inclusion of new hypotheses after the interim analysis, in this case corresponding to new subgroups, was also raised as an extension to seamless designs by Hommel [24]. These are topics for possible future investigation.

Exact bivariate methods [13] could also be investigated to replace the multiplicity correction methods for deriving p -values for the intersection hypothesis H_0^{FS} of no treatment effect on OS in either the full population or the subgroup. The Hochberg procedure can introduce some slight conservatism by not taking account of the correlation between the populations being tested under each hypothesis, although this could be accounted for by using Dunnett's method [13]. Further, the formula presented to calculate an adjusted p -value from the Hochberg method could be adapted to incorporate an uneven split of the Type-I error probability. Posch *et al.* [25] consider various combination tests and multiplicity correction procedures, as well as introducing confidence intervals. The Hochberg procedure was shown to perform well when compared with 'splitting the alpha' type methods [9].

8. CONCLUSIONS

We have shown how practical design aspects for oncology trials can be applied to allow adaptive seamless methods to be used in a subpopulation selection problem. Novel features such as the co-primary population, the use of an intermediate endpoint for earlier decision-making, clearly defined decision rules and the recruitment of larger numbers of subgroup patients, if required, have been added. Simulations show that, in agreement with theory, these modifications have not inflated the Type-I error rate and that the design performs well in terms of the final conclusions that would be reached in various situations.

If a pre-defined hypothesis for a predictive biomarker exists, this approach could enable effective medicines to be made available more rapidly to the most appropriate patient population.

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